

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 1: Cuttings, Tubers and Lonely Trees: Why Do Plants Reproduce So Differently?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	B	D	C	A	B

USE AFTER: PREDICT PHASE — Phase 1 Predict: farm scene chart of cuttings versus seeds and initial model drawing

Q1. Farmers in a Kenyan community plant pieces of cassava stem and sweet potato vines to get new plants, but they buy maize and bean seed before the long rains. Which statement best describes what the farmers are doing?

- A. All crops must be planted as seeds, so the cassava stem must contain hidden seeds
- B. Maize and beans are grown from cuttings, while cassava is grown from seeds
- C. Cassava and sweet potato need a flower on the cutting before they can grow
- D. Some crops are grown from pieces of stem or vine without seeds, while others are grown from seeds**

Correct answer: D. The farm scene shows cassava, sugarcane and sweet potato growing from cuttings with no seed, while maize and beans are planted as seeds. The wrong choices reflect the preconception that all plants need seeds or flowers.

USE AFTER: OBSERVE PHASE — Phase 2 Observe: specimen table, ginger rhizome with nodes and buds

Q2. A group examines a ginger piece with a hand lens. It is a thick stem that grows along or just under the soil. It has nodes with small buds along it. Which structure type is it?

- A. Bulb
- B. Runner
- C. Tuber
- D. Rhizome**

Correct answer: D. A ginger piece is a rhizome: a thick underground stem with nodes and buds from which new shoots grow.

USE AFTER: OBSERVE PHASE — Phase 2 Observe: onion bulb cut open by the teacher

Q3. A teacher cuts an onion bulb in half from top to bottom. What will learners see inside?

- A. A hard seed coat with a tiny embryo inside
- B. A short stem at the base with thick, fleshy leaves that store food**
- C. Many thin roots joined to a fruit
- D. A flower that has not yet opened

Correct answer: B. An onion bulb is a short stem (stem plate) surrounded by fleshy leaves that store food. It is not a seed.

USE AFTER: OBSERVE PHASE — Phase 2 Observe: bryophyllum leaf with plantlets on the margin

Q4. A bryophyllum leaf has tiny plantlets growing on it. Where on the leaf do these plantlets grow?

- A. From a flower on the leaf surface
- B. Only from the leaf stalk buried in the soil
- C. From seeds in the middle of the leaf
- D. From the edge (margin) of the leaf**

Correct answer: D. Bryophyllum plantlets form along the leaf margin. They grow without seeds or flowers and can fall off and root in the soil.

USE AFTER: OBSERVE PHASE — Phase 2 Observe: teacher question 'How is a spore different from a seed?' with bread mould or fern picture

Q5. Ferns and mushrooms reproduce by spores (sporulation). Which statement correctly tells a spore apart from a seed?

- A. A spore contains an embryo and a food store, while a seed is a single cell
- B. A spore and a seed are the same thing with different names
- C. A spore is a single cell with no stored food, while a seed contains an embryo and a food store**
- D. A spore forms only inside flowers, while a seed forms on leaves

Correct answer: C. Spores are single cells with no stored food. A seed contains an embryo together with stored food. This difference was discussed when the teacher showed bread mould or a fern frond.

USE AFTER: EXPLAIN PHASE — Phase 3 Explain: Think-Pair-Share on advantages and disadvantages of growing cassava from cuttings

Q6. A farmer grows a whole field of cassava from cuttings taken from one parent plant. A disease called cassava mosaic attacks, and the parent plant is weak against it. What is most likely to happen to the plants in the field?

- A. Because the plants are identical to the parent, many of them may also be weak against the disease**
- B. Each plant is different, so some will always resist the disease
- C. Cuttings do not carry diseases, so the field will be safe
- D. The new plants will become stronger because they have no seeds

Correct answer: A. Growing from cuttings of one parent gives offspring identical to that parent, so they share its weakness. This is a disadvantage of asexual reproduction.

USE AFTER: DQB PHASE — Phase 4 DQB: questions about the lone avocado or pawpaw posted under 'With Seeds'

Q7. A lone avocado tree in a compound is covered in flowers but never sets fruit. After today's lesson on cuttings, tubers, bulbs, rhizomes and runners, what is the best statement about this tree?

- A. It is explained: avocado trees have no bulbs, so they cannot make fruit
- B. It is not explained yet: it is a question for the Driving Question Board and the next lesson on flowers**
- C. It is explained: flowers on a tree always make fruit on their own
- D. It is explained: the tree reproduces by runners, so it does not need fruit

Correct answer: B. Today's lesson shows how plants make new plants without seeds. It does not explain why a flowering tree gives no fruit, so this stays an open question on the board for the next lesson.